Given a density ρ, we may identify a collection of functions g for which the integral with respect to ρ is bounded, i.e. for which we can define the inner product $\langle g, \rho \rangle = \int_{\mathcal{D}} g(z)\rho(z)\mathrm{d}\omega$, and thus we can define

$$\mathrm{Av}_\rho(g) \stackrel{\mathrm{def}}{=} \frac{1}{\int_{\mathcal{D}} \rho(z)\mathrm{d}\omega} \langle g, \rho \rangle = \int_{\mathcal{D}} g\hat{\rho}\mathrm{d}\omega.$$

We term this the *spatial average* of the function g with respect to the probability measure associated to ρ.

5.2.2 Liouville Equation

We now use the concept of a smooth probability density to discuss the change in concentration of points present in a given small volume in phase space.

Let $\rho(z, t)$ be the probability density associated to a collection of phase points moving under a vector field $\boldsymbol{f}$ on the phase space $\mathbb{R}^m$. Let $\mathcal{A}$ be a suitable region of volume V within the phase space (say a rectangular m-dimensional box). The fraction ν of phase points present in $\mathcal{A}$ at time t is the integral of ρ over $\mathcal{A}$:

$$\nu(t) = \int_{\mathcal{A}} \rho(z, t)\mathrm{d}\omega,$$

which may be viewed as a smooth function of time. On the one hand

$$\frac{\mathrm{d}}{\mathrm{d}t}\nu = \int_{\mathcal{A}} \frac{\partial \rho(z, t)}{\partial t}\mathrm{d}\omega,$$

while, on the other hand, this same rate of change may be written as the negative of the particle flux over the boundary of $\mathcal{A}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\nu = -\int_{\partial\mathcal{A}} \boldsymbol{f} \cdot \boldsymbol{n}\rho(z, t)\mathrm{d}\sigma,$$

where $\boldsymbol{n}(z)$ is the unit outward normal to the surface $\partial\mathcal{A}$ at the point z and $\mathrm{d}\sigma$ is the surface element. Using the divergence theorem, we may rewrite the latter quantity as a volume integral:

$$\frac{\mathrm{d}}{\mathrm{d}t}\nu = -\int_{\mathcal{A}} \nabla \cdot (\boldsymbol{f}(z)\rho(z, t))\mathrm{d}\omega.$$